Fine-tuning Multimodal Transformers on Edge: A Parallel Split Learning Approach

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01 Context & approach

The challenge

Multimodal transformers are costly to fine-tune on edge devices. Even parameter-efficient federated methods execute the backbone locally. *MPSL* keeps lightweight image, audio and text tokenizers on clients while a shared server encoder performs the expensive computation.

Our contribution

Extend parallel split learning to paired modalities with one shared encoder.
Aggregate client losses for one server backward pass while raw labels stay local.
Evaluate task performance and client cost across seven multimodal datasets.

1 Tokenize and fuse

Clients tokenize each modality. Early fusion joins client tokens; late fusion joins representations after server encoding.

2 Train through one server

Return predictions for local loss computation. Aggregate losses, backpropagate once, then return cut-layer gradients for tokenizer updates.

3 Construct the evaluation model

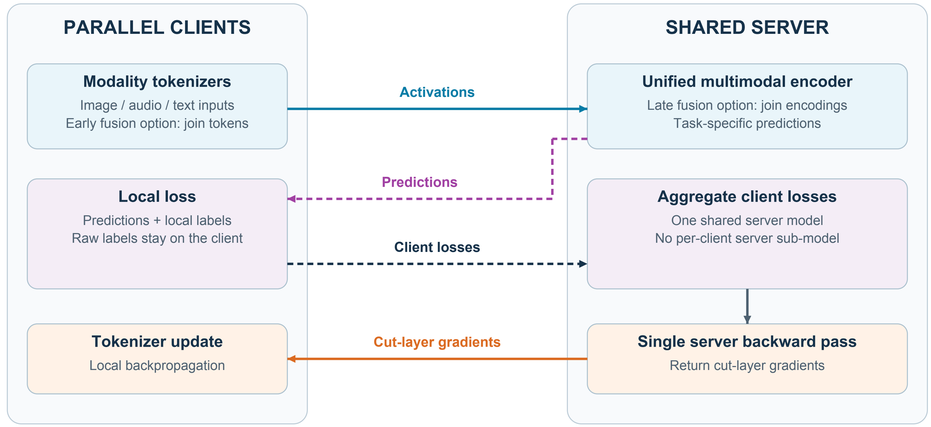
After training, average client tokenizers and join them with the shared server encoder.

Scope

Client savings shift computation to the server; classification performance varies by task. Retrieval trails centralized training; missing-modality robustness is untested.

02 Method

Keep multimodal tokenization on clients and move transformer fine-tuning to a shared server.



Core formulation

$$\mathcal{L}_S = \sum_{n \in \mathcal{N}} \frac{|\mathcal{B}_n|}{|\mathcal{B}|} \mathcal{L}_{C_n}$$

L_Cn: local client loss; B_n: client batch; B: combined batch. Minimize this batch-weighted loss in one server backward pass.

Paper

Code

03 Results

0.1 GFLOPs
average client compute
Per input, five classification tasks; FedAvg: 25.8.

58.2 ± 0.9%
MELD accuracy
100 clients; FedAvg: 48.9 ± 0.0%.

Client efficiency with task-dependent accuracy

Method	COCO-QA	UCF101	MELD	Client GFLOPs
Centralized	61.7 ± 0.1	91.3 ± 0.2	61.4 ± 0.4	25.8
FedAvg	53.5 ± 0.2	89.8 ± 0.9	48.9 ± 0.0	25.8
FedCLIP	13.1 ± 2.6	56.4 ± 1.2	50.7 ± 0.7	25.8
MPSL (Ours)	**46.2 ± 0.5**	**89.3 ± 1.0**	**58.2 ± 0.9**	**0.1**

Accuracy (%), reported ± values, 3 seeds. Distributed methods: 100 clients, Dir(0.1); Meta-Transformer ViT-B/16, last 6 blocks trained. Client GFLOPs/input averaged across five classification tasks.

Image-to-text retrieval remains challenging

Method	MS-COCO R@1	Flickr30K R@1	Client GFLOPs
Centralized	56.8 ± 2.4	66.8 ± 5.3	8.65
FedAvg	1.5 ± 0.1	11.2 ± 1.7	8.65
FedCLIP	0.1 ± 0.0	0.1 ± 0.0	8.65
MPSL (Ours)	**21.1 ± 0.8**	**26.0 ± 0.5**	**0.1**

Image-to-text recall@1 (%), reported ± values, 3 seeds. Distributed methods: 100 clients; full ViT-B/16 encoder fine-tuning. GFLOPs are per client input.

Takeaway

MPSL reduces average client compute to **0.1 GFLOPs per input**, but classification gains depend on the task: MELD improves over FedAvg while COCO-QA declines. Retrieval exceeds the tested federated baselines yet remains well below centralized training.